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The Metadata on your Code and Text – working with Git and GitHub –

(presentation was prepared with the help of ChatGPT)

TIPPECC

Franziska Zander (Krüger NP 9.03.2026)

Agenda

- Why version Control?
- Git vs GitHub
- Use Cases
- Common Workflow
- Git Concepts
- Git Basics
- Practise
- Summary & Questions

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Introduction

- Who is using Git?
- Who is using GitHub, Gitlab, ...?
- How do you save your work? (code, text, tasks, ...)

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Why version control?

- doc_final.txt
- doc_final_final.txt
- doc_final2.txt
- doc_final_final_2.txt
- doc_final_FZ_final.txt

What is the latest version?

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Why version control?

- History (go back, restore, compare, who changed and when, ...)
- Safety (experiments, ...)
- Collaboration (no overwrites, no conflicts, latest version known, ...)
- Traceability (author, time, why, ...)
- Backup
- Documentation (issues, ...)
- Workflows (ci /cd, ...)
- Structured releases (tag releases, ...)
- ...

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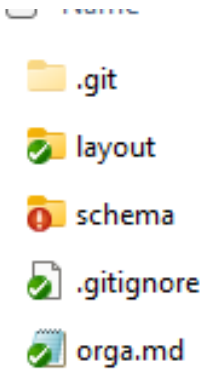
History of Git & GitHub

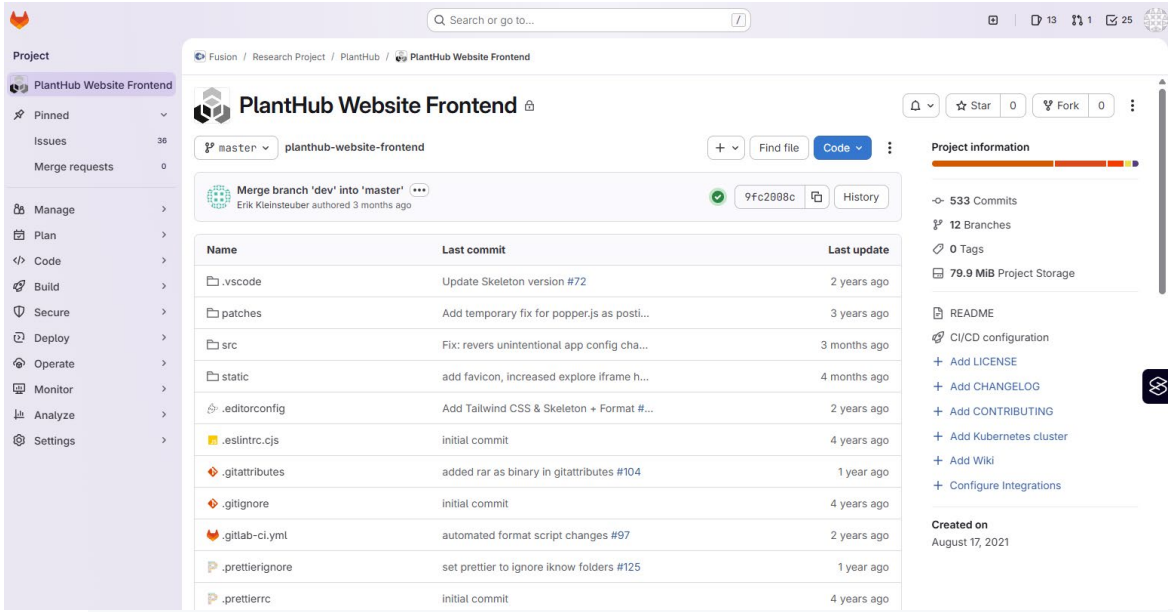
- In 2005 BitKeeper withdraw free licence for Linux kernel development
- Linus Torvalds (inventor of Linux) invented it within few weeks
- Naming:
  - git - british English slang for an unpleasant or silly person
  - *"I'm an egotistical bastard, and I name all my projects after myself. First 'Linux', now 'git'."* (https://archive.kernel.org/oldwiki/git.wiki.kernel.org/index.php/GitFaq.html#Why_the_.27Git.27_name.3F)
- In 2008 GitHub was founded
- 2010 -2015 Git gaining ground
- 2018 GitHub was bought by Microsoft
- Today: Git is the dominant version control system worldwide.

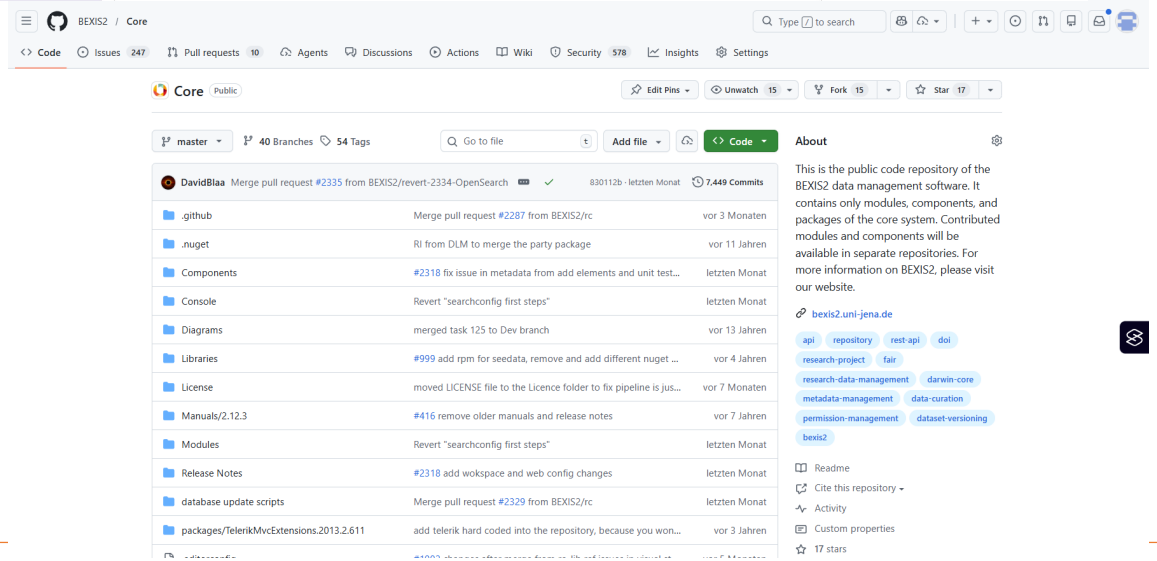
Git vs. GitHub/GitLab/ ...

- Git
  - Tool
  - Runs locally
  - Tracks changes
  - Works offline
- GitHub & co
  - Platform
  - Hosts Git repositories
  - Adds collaboration:
    - Issues
    - Permissions
    - Reviews
    - Pull requests
    - ...

https://about.gitlab.com/
https://github.com/







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Use Case 1 - Paper

File Edit Insert View Format Help DBSPEKTRUM-26-dbis-groups-jena Share Upgrade

Code Editor Visual Editor Recompile 1 / 14 63%

dbspektrum-26-dbis-groups-submitted > dbspektrum-26-dbis-groups-par

```
Across these strands, the group pursues a human-
centric engineering perspective, aligning rigor
semantic modeling with practical system architectures
to support transparency, flexibility, and reuse in
dynamic resource ecosystems.


\subsection*{Application Areas: Research and Research
Data}

\subsubsubsection*{Platform development: BEXIS2}
    The development of the research data management
    platform BEXIS2\footnote{\url{https://bexis2.uni-
    jena.de/}} \cite{Chamanara2021BEXIS2} began in
    2007 and has undergone substantial evolution over
    time. Today, BEXIS2 is utilized in over 20 mostly
    large-scale research projects. The platform
    supports the collection and curation of both
    tabular and non-tabular research data, together
    with their associated metadata. By enabling the
    assignment of Digital Object Identifiers (DOIs),
    BEXIS2 accommodates contemporary data publication
    requirements and workflows. To meet the diverse
    needs of research projects, the system allows
    project-specific metadata configurations and the
    concurrent use of multiple metadata schemas.
    Furthermore, BEXIS2 offers flexible mechanisms
    for defining optional, nested metadata structures
    and controlled vocabularies, enabling metadata
    models to be tailored to discipline-specific
    requirements.
```

The Chairs for Databases and Information Systems at the Friedrich-Schiller-Universität Jena

Waqas Ahmed, Tarek Al Mustafa, Sarah T. Bachinger, Maik Fröbe, Matthias Hagen, Birgitta König-Ries, Marianne Mauch, Jan Heinrich Merker, Divyasha Sunil Naik, Andreas Ostrowski, Wilhelm Pertsch, Pascal Scherreiks, Ferdinand Schlatt, Franziska Zander, Ines Zelch

Friedrich-Schiller-Universität Jena, Jena, Germany.

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Abstract

At the Friedrich-Schiller-Universität Jena, the Heinz-Nixdorf Chair for Distributed Information Systems and the Chair for Databases and Information Systems are led by Matthias Hagen since 2004 and 2022, respectively. The two research groups focus on information integration, semantic web technologies, information retrieval, natural language processing. The developed technologies are applied in various domains such as digitization, argument mining, web search, conversational systems, retrieval-augmented generation.

Keywords: Jena, information integration, information retrieval, natural language processing

1 Origins and Evolution of Database and Information Systems Research in Jena

Databases and Information Systems have played a central role in the work of the Faculty of Mathematics and Computer Science at the Friedrich-Schiller-Universität Jena since its foundation following the German reunification. From 1991, Albrecht Blaser, a former IBMer, represented the topics in Jena. After a brief interim chairmanship by Volkert Bros... chair in 1995. F... cused on core ... work addressed ... DBMSs, includ... partitioning, da... design, and per... tures. In the ear... toward workflow... information syst... performance da... tributed usage ...

1

```
\section{FUSION -- Heinz Nixdorf Chair for Distributed
Information Systems}
The FUSION group started its work quite exactly twenty
years ago. From the outset, the~~very beginning, the~~
central theme of its work has been~~is~~ the transparent,
integrated usage of resources in open, heterogeneous,
and dynamic environments, hence the abbreviation FUSION
(FUnctionality Sharing In Open eNvironments). While the
application areas we focus on have evolved over time,
the main methodological focus has stayed with Semantic
Web approaches and strong user integration.

\subsection*{Approaches}
Methodologically, the FUSION group focuses on Semantic
Web-based integration and access to distributed
resources in open and heterogeneous environments.
Central to our work are ontology-based representations,
semantic annotation, and knowledge graph construction,
which provide a shared conceptual layer for integrating
diverse data and service resources and for enabling
```

Sun, 25th Jan 26

25th January, 1:11 pm
Edited dbspektrum-26-dbis-groups-submitted/dbspektrum-26-dbis-groups-part1.tex
Edited dbspektrum-26-dbis-groups-submitted/dbspektrum-26-dbis-groups-part3.tex
birgitta.koenig-ries

25th January, 1:06 pm
Edited dbspektrum-26-dbis-groups-submitted/dbspektrum-26-dbis-groups-part1.tex

File outline
- FUSION -- Heinz Ni...
  - Approaches
  - Application Are...
    - Platform d...
    - Research d...
    - Contributin...
    - Bringing se...
    - Combining ...
    - Exploring a...

https://www.overleaf.com/
(with pro you can link to a local repo & see full history)

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Use Case 2 – Code (Issues + Commit)

TIPPECC / tippecc.github.io

Issues 17 | Pull requests | Zenhub | Agents | Discussions | Actions | Projects 1 | Wiki | Security 35 | Insi

Remove hard-coded settings and replace with backend call #135

Closed

geofranzi opened on 5. Dez. 2025

see geofranzi/geoportal#116

Create sub-issue

schreifab on 8. Jan. · edited by schreifab

2 main challenges remaining:

- [x] the sum symbol is no correctly provided via the API
- [x] FoldertypeChooser is loaded before the folder types are returned from the API

schreifab self-assigned this on 9. Jan.

schreifab added 2 commits that reference this issue on 9. Jan.

```
add store for folderytpes and fetch from api #135
Bug Fix: Bbox is an array now #135
```

TIPPECC / tippecc.github.io

Type / to search

Code | Issues 17 | Pull requests | Zenhub | Agents | Discussions | Actions | Projects 1 | Wiki | Security 35 | Insights | Settings

Commit 708c9ea

Browse files

schreifab committed on Jan 12 · 3 / 3

```
Bug Fix: Bbox is an array now #135
```

master

1 parent 464e4b8 commit 708c9ea

Filter files...

src/lib/tempresults

foldertype_chooser.svelte

1 file changed +3 -3 lines changed

Search within code

.../tempresults/foldertype_chooser.svelte +3 -3

```
@@ -24,7 +24,7 @@
24  24          header_regex: string;
25  25          lineage?: string;
26  26          status: string;
27     -        bbox?: string;
   27  +        bbox?: [number, number, number, number];
28  28      }[] = $folder_types;
29  29
30  30      const foldertypes_full = foldertypes;
@@ -137,9 +137,9 @@
137 137                     bind:boundingBox={bbox}
138 138                     bind:startDate
139 139                     bind:endDate
140    -                    startingExtent={JSON.parse(
   140 +                    startingExtent={
```

https://github.com/TIPPECC/tippecc.github.io/issues/135

Use Case 3 – Scrum (Tasks distribution + Review)

BEXIS2 / Projects / BEXIS2

Type / to search

BEXIS2

Add status update | Insights | Workflows 2

Scrum-Backlog | No Status | Issues Overview | By milestones - review | by milestones - open | View 13 | Priority | By priority | By Size | By Status | Feature Request | Improvme

-status:"Drafts","Abandoned" 343 View

Backlog 10

s & Seed data Units, Ext Links, ...
edium | Type: Feature Request

on Timeout in Svelte

d external Link
Medium
quest | maintenance

ta structure: Duplicate IDs

In progress 20

Core #1111
automatic integration into metadata of DOI after minting

Core #1621
Removal of PostSharp Dependency

Core #2190
Tags: Improve default text

Core #2068
Use of ORCID and ROR IDs in the user profile and in the DOI proxy
Type: Improvement Proposal

Core #1992
Data Curation Workflow
Type: EPIC

In review 8

Core #2053
Dynamic loading of components for edit

Core #2186
DIM Seed Data Update

Core #2206
Entity Extension: Add selection of extensions to entity templates
Type: Feature Request

Core #2209
Entity Extension: Allow to add extensions on edit page
Type: Feature Request

Core #2233
Entity Extension: Hide Options from EntityTemplate Form

Done 30

Core #2201
Species Mapping: Create Species Mapping Module (SMM)
Urgent | Medium | Type: Feature Request | feature request

Core #2294
Curation Tool: Feature permissions missing in update script

Core #2363
Internal Metadata View: Links and Data Structure not shown
Urgent

Core #2364
PUM - Bugs

Hold 8
waiting for something

Core #892
Update Attachment API
Low | Small | maintenance | Type: Feature Request | feature request

Core #2130
DIM: API Metadata Get JSON Schema no choises

Core #936
UI for Versioning: visibility and accessibility on public and internal datasets

Core #1308
Email Header unification

https://github.com/orgs/BEXIS2/projects/8/views/1 (private)

Use Case 4 – Task Organization

https://github.com/bexis/Data-Curation/issues (private)

github.com/bexis/Data-Curation/issues?q=is%3Aissue%20state%3Aclosed

bexis / Data-Curation

Code | Issues 144 | Pull requests | Agents | Actions | Projects 3 | Wiki | Security | Insights | Settings

Monthly tasks
#123 · CFuerstenau opened on Mar 14, 2022 · 6

R-Scripts Check
#937 · CFuerstenau opened on Nov 21, 2024 · 2

Data mobilization
#1009 · CFuerstenau opened on Jan 15, 2025

is:issue state:closed

Open 144 | Closed 1.038

- **Dataset 32318 - Akshda** Curation: metadata update | Curatior
  #1334 · by CFuerstenau was closed 2 weeks ago · 2026/03
- **Publication 31999 - Schweizer et al.** Publication: submitted
  #1319 · by PaulaTri was closed on Jan 15
- **Dataset 32306 - Falk-Rudhard S.** Curation: metadata update
  #1317 · by CFuerstenau was closed 3 weeks ago · 2026/02
- **Dataset 32297 - Vidisha B.** Curation: metadata update | Data
  #1313 · by CFuerstenau was closed on Jan 14 · 2026/02
- **Dataset 32296 - Vidisha B.** Curation: metadata update | Cura
  #1312 · by CFuerstenau was closed 3 days ago · 2026/02
- **Dataset 32295 - Sara B.** Curation: metadata update
  #1311 · by CFuerstenau was closed 2 weeks ago · 2026/02

github.com/bexis/Data-Curation/issues/1312

bexis / Data-Curation

Code | Issues 144 | Pull requests | Zenhub | Agents | Actions | Projects 3 | Wiki | Security | Insights | Settings

Dataset 32296 - Vidisha B. #1312

Edit | New issue

Closed

CFuerstenau opened on Dec 29, 2025 · edited by CFuerstenau | Edits

Metadata

- [x] Title:
  - [x] Year
  - [x] Location
  - [x] Information about data
  - [x] your data contains 100 plots. Please change this in the title
  - [x] please do not use BE specific abbreviation in the title like EP (e.g., Pollen quantification from Ranunculus acris and Trifolium pratense from 100 grassland plots in 2023-24)
- [x] Abstract:
  - [x] Timeframe
  - [x] please add 2 or 3 sentences about the motivation or background of the study
  - [x] Location: please add following information ("in the grassland plots" from three German regions (Schwäbische Alb, Hainich-Dün, Schorfheide-Chorin)
  - [x] Information about which data the dataset contains
- [x] Contact Person:
- [x] Number of plots: GP/EP = 100, MIP = 52, VIP = 15
- [x] Study design:

Assignees: CFuerstenau
Assign to Copilot

Labels: Curation: metadata update | Curation: variable template/unit changed | Dataset: doi

Type: No type

Projects: Data Curation: Dataset quality check — Status: Done

Milestone: 2026/02 — Due by February 15, 2026, 42% complete

Relationships

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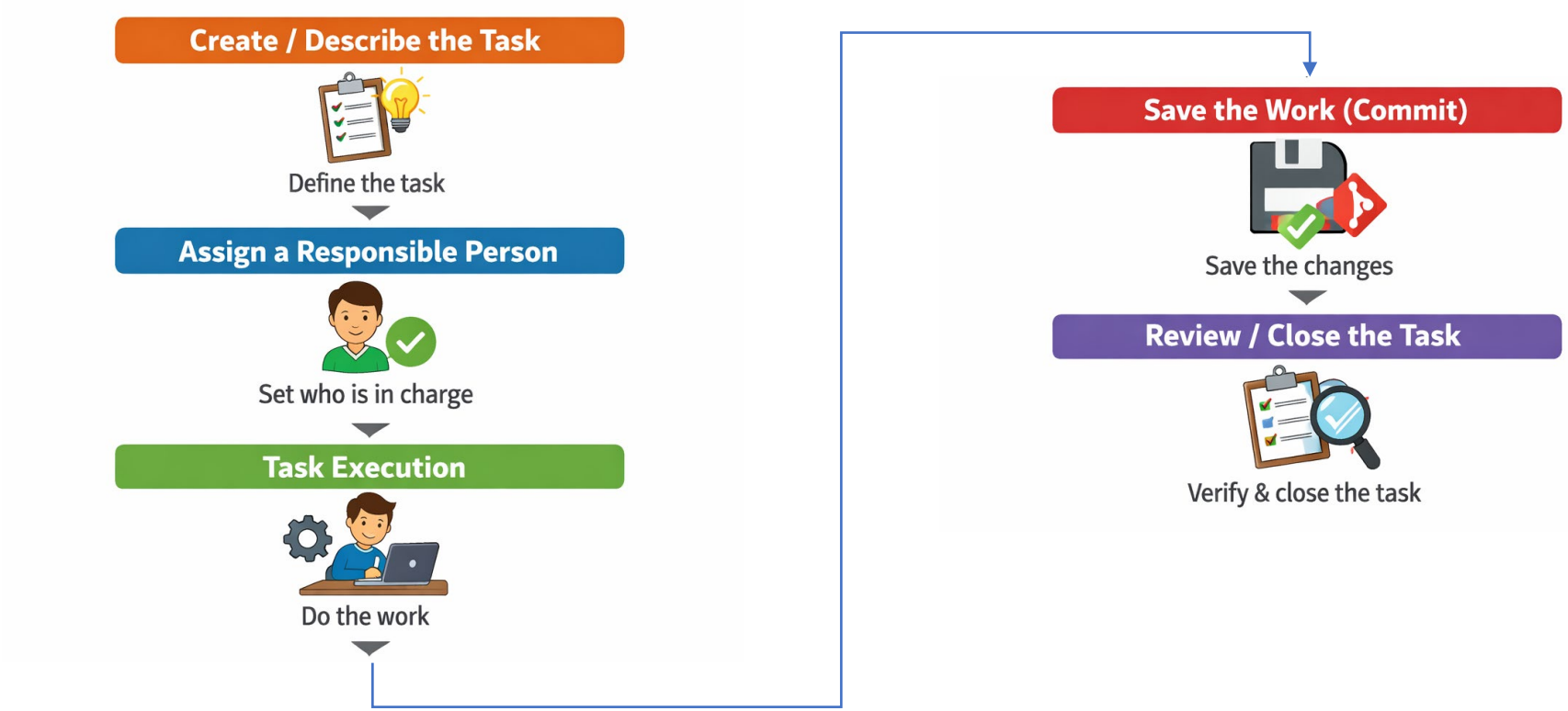
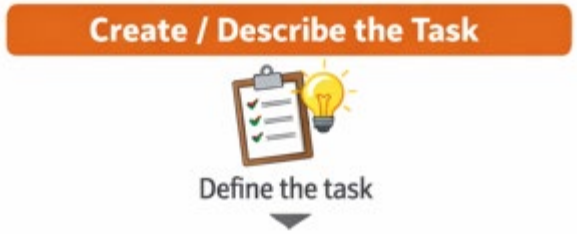
University of Botswana

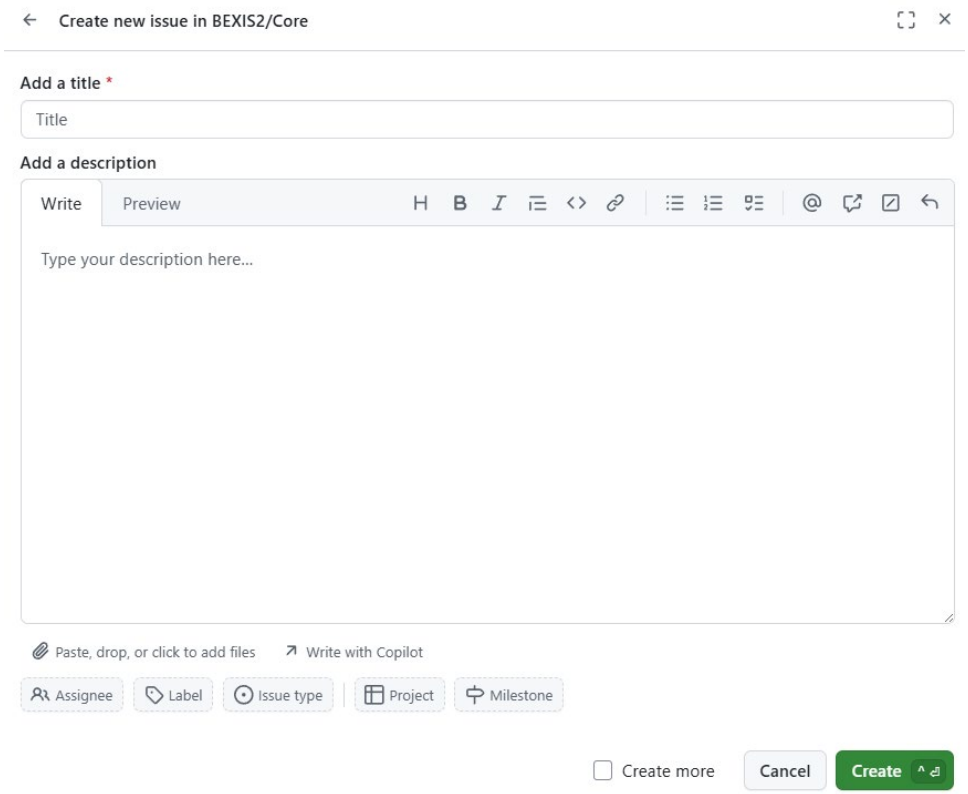
Common Workflow (... in software development)



Task Description: Write a good issue (1/3)

- What types of issues (structured description of work)?
  - A bug
  - A feature request
  - A task
  - A discussion
  - A technical improvement
  - ....





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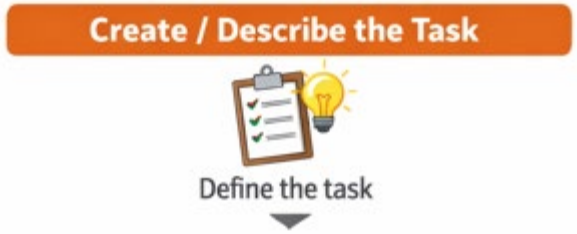
Task Description: Write a good issue (2/3)



- 1. Clear Title
  - No: Fix Login
  - Yes: Login fails when password contains special characters
- 2. Context (What/Why/Who …)
  - Users cannot log in if their password contains characters like “#” or “&”.
  - This blocks approximately 15% of users based on current feedback.
- 3. Expected vs Current Behavior (for bugs)
  - Current behavior: User receives “Invalid credentials” error.
  - Expected behavior: User should be able to log in with any valid password.

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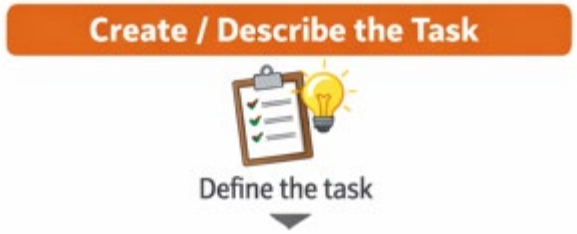
Task Description: Write a good issue (3/3)



- 4. Acceptance Criteria (when done?)
  - ✓ Login works with special characters
  - ✓ Error messages are clear
  - ✓ Existing users are unaffected
  - ✓ Automated tests updated

- 5. Additional Information
  - Screenshots
  - Links
  - Environment details
  - Related issues
  - Technical hints (Browser …)

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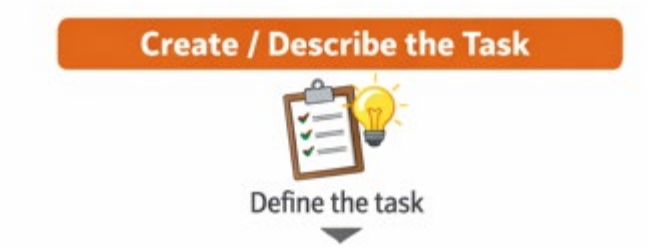
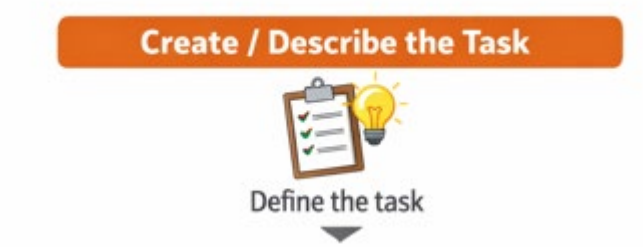
Task Description: Summary



- Be precise
  - No: Make it better
- Keep one issue = one topic
  - Do not mix bugs and features
- Write for someone who was not in the meeting

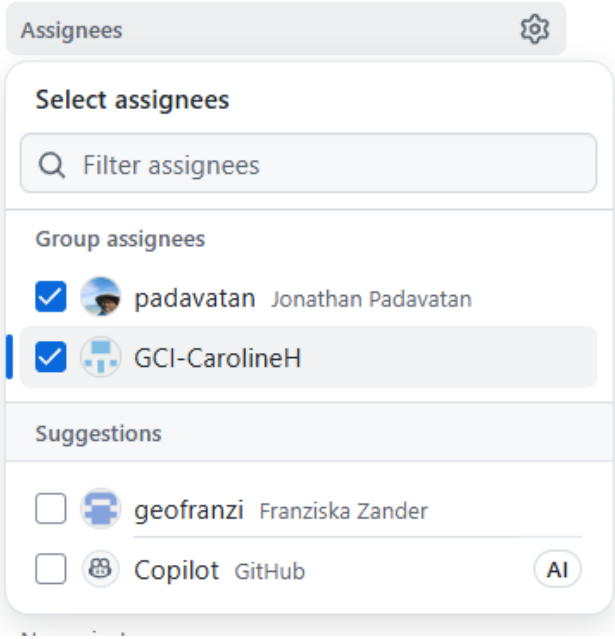
Issue Templates - Example

- https://github.com/BEXIS2/Core
  - -> Issues -> New Issue -> Bug Report



Task Assignment / Planning / ....

- Assign tasks



- Add milestone / priority /...
  - Milestone: https://github.com/BEXIS2/Core/milestones

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Task execution

It works ☺

- Go home?

- … No! ;-)

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Save & Document the Work

Commits

History for CSG_Preprocessing / Install_and_Maintain on `main`

Commits on Feb 19, 2026

Add gateway temp folder and docker install for GraphDB #3 #5
geofranzi committed 2 days ago

Add graphDB import and related libs #5
geofranzi committed 2 days ago

Commits on Feb 18, 2026

Add preprocessing installation instructions and code #12
geofranzi committed 3 days ago

Commits on Feb 15, 2026

Add Server installation & maintenance docu #9
geofranzi committed last week

End of commit history for this file

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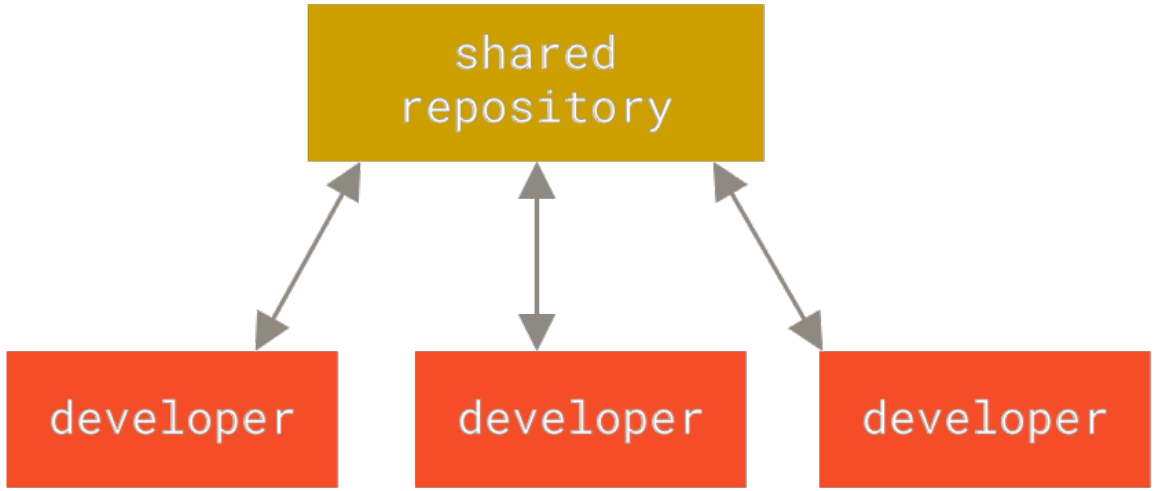
Core Git Concepts (1/2)

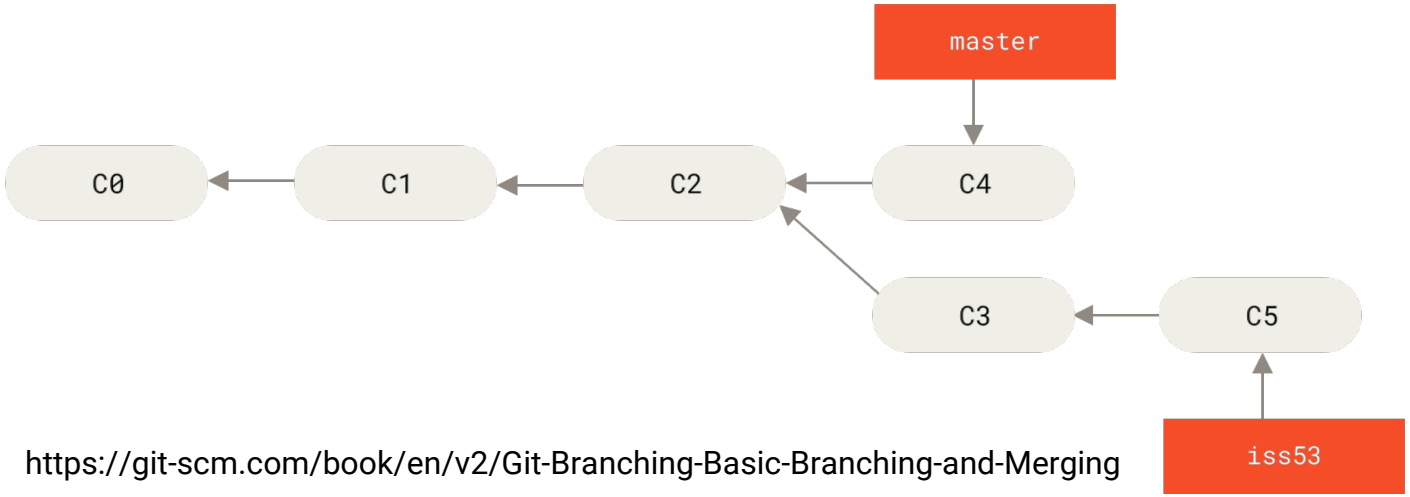
- Repository
  - Project folder + history
- Commit
  - Snapshot of changes
  - Has
    - Author
    - Message
    - Timestamp
  - Best: small, meaningful commits



https://git-scm.com/book/en/v2/Getting-Started-About-Version-Control

Core Git Concepts (2/2)

- Branch
  - Parallel timeline
  - Safe experimentation
  - main / mainline as stable base



https://git-scm.com/book/en/v2/Git-Branching-Basic-Branching-and-Merging

- Remote vs local
  - Local repo
  - GitHub repo push / pull

```
(venv_3_12_preprocessing) root@sapphire:/opt/CSG_Preprocessing# git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   .gitignore
        modified:   Install_and_Maintain/Backend_CSG.md
        modified:   Install_and_Maintain/Frontend_CSG.md
        modified:   Install_and_Maintain/Processing_CSG.md
```

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```
🌍 Remote Repository (GitHub / Server)
            |
            |  git clone (first time)
            ▼
   💻 Your Local Repository
            |
     (edit your files)
            |
     git status  🔍
            |
     git add     ➕
            |
     git commit  💾
            |
            |  git push  🚀
            └──────────────► Upload changes


◄──────────────────
      git pull  🔄
  (get others' updates)
```

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Basic Git Commands

- git clone (creates a local copy of a remote repository.)
  - git clone https://github.com/class/project.git
- git checkout (switches to another branch or restores files.)
  - git checkout class1
- git status (shows the current state of your working directory.)
- git add (stages changes for the next commit.)
  - git add slides.txt
- git commit (saves the staged changes as a new version in the project history.)
  - git commit -m "Added introduction slide"
- git push (uploads your local commits to the remote repository.)
- git pull (downloads new changes from the remote repository and merges them into your current branch.)

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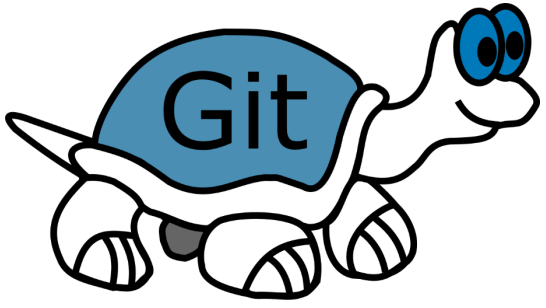
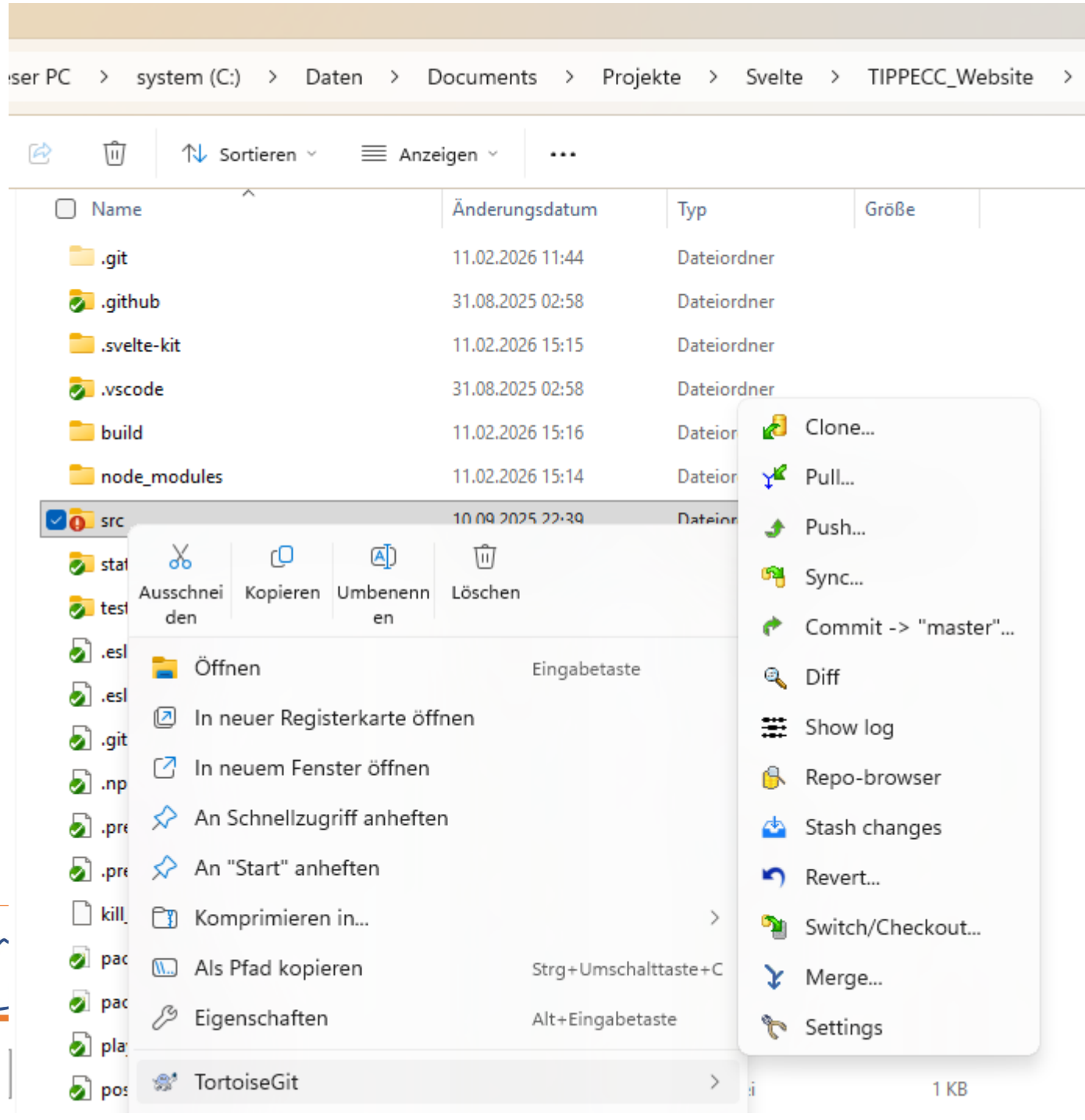
Git: Tools

- https://tortoisegit.org/



- https://git-scm.com/install/windows
- (Windows, macOS, Linux)



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Git: Tools - Example Visual Studio Code

File Edit Selection View Go Run Terminal Help

SOURCE CONTROL

CHANGES

Message (Ctrl+Enter to commit on "...

Commit

Merge Changes 1

package-lock.json

Staged Changes 1

package.json M

Changes 57

src

docs

components

CodeContainer.svelte M

Navigation.svelte M

lib

components

CodeEditor

CodeEditor.svelte M

Facets

Facets.svelte 1, M

page

Table

ColumnsMenu.svelte M

TableContent.svelte M

TablePagination.svelte M

TablePaginationServer.svelte M

css

themes

theme-bexis2.css M

theme-bexis2.css_old U

core.ui.postcss M

shims\skeleton

index.ts M

popup.ts M

GRAPH

GITLENS

core.ui.postcss M | Facets.svelte (Working Tree) 1, M | theme-bexis2.css_old U

src > lib > components > Facets > Facets.svelte > script

```
<script lang="ts">
    import { createEventDispatcher } from 'sv
    import type { Writable } from 'svelte/sto

    import ShowMore from './ShowMore.svelte';

    import type { FacetGroup, SelectedFacetGr

    export let groupSelection = false;
    export let groups: Writable<FacetGroup[]>
    export let showAll = false;
    export let open = false;

    export const showMore = (groupName: strin
        const group: SelectedFacetGroup = sel

        dispatch('showMoreOpenChange', {
            group: group.name,
            open: true
        });

        modalStore.trigger({
            type: 'component',
            title: `${group.displayName}`,
            component: {
                ref: ShowMore,
                props: {
                    group,
                    handleApply,
                    handleCancel
                }
            }
```

```
<script lang
    import {
    import t

    import S
+   import {
+   import {
    import t

    export l
    export l
    export l
    export l
+   export l
+   export l

    export c
        cons

        disp

        });

        moda
```

PROBLEMS 5 SPELL CHECKER 4 DEBUG CONSOLE TERMINAL PORTS GITLENS

TERMINAL

Executing task: npm run dev

upgrade-svelte-5*+ Launchpad 1 0 4 1

CHANGES

Message (Ctrl+Enter to commit on "...

Commit

Changes

Install_and_Maintain

Backend_CSG.md

Frontend_CSG.md

Processing_CSG.md

GRAPH Auto

Add details for calc_indicators.... main

Update README for provenance and test r

Update README with guidelines and proce

Initial commit Franziska Zander

nstall_and_Maintain > Backend_CSG.md >

View as List

View & Sort

Pull

Push

Clone

Checkout to...

Fetch

Compose Commits (Preview)...

Generate Commit Message with GitLens

Add Co-authors...

Commit

Changes

Pull, Push

Branch

Remote

Stash

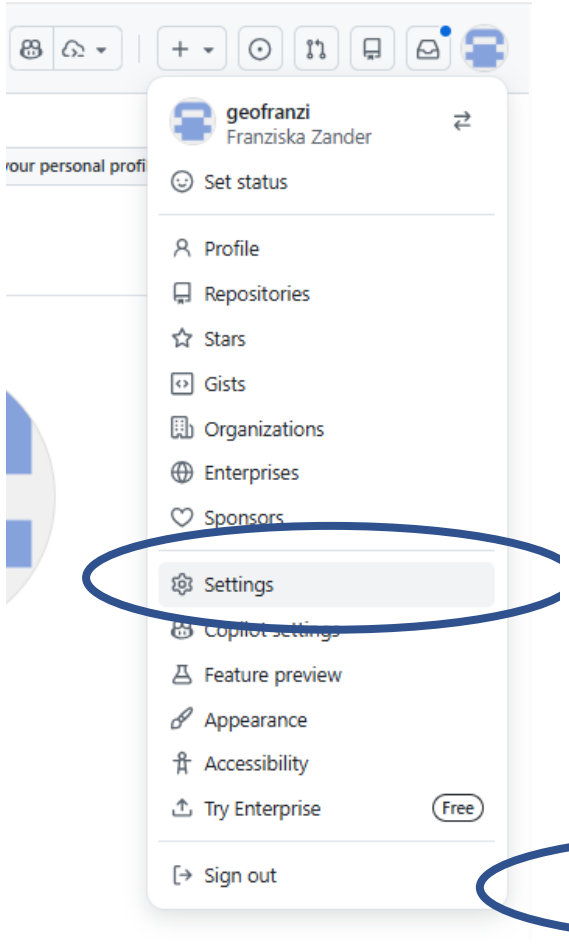
Tags

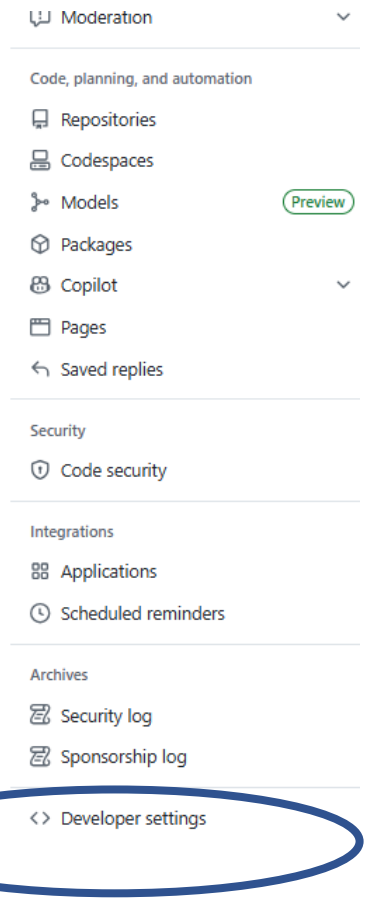
Show Commit Graph

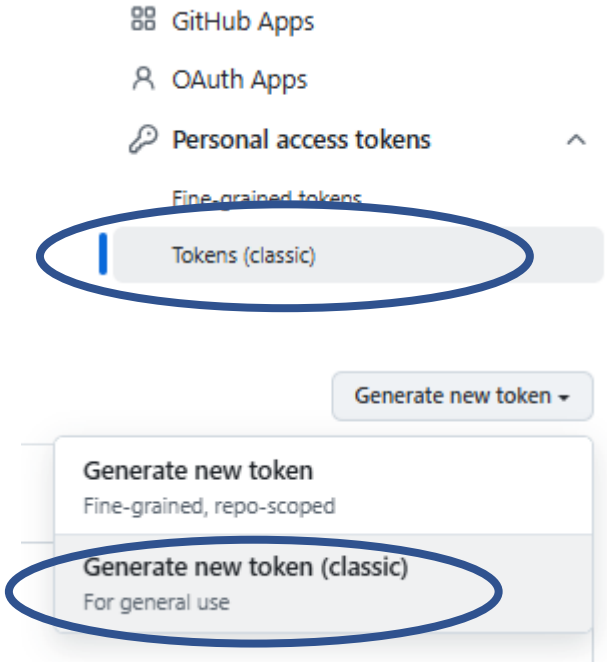
Visualize Repository History

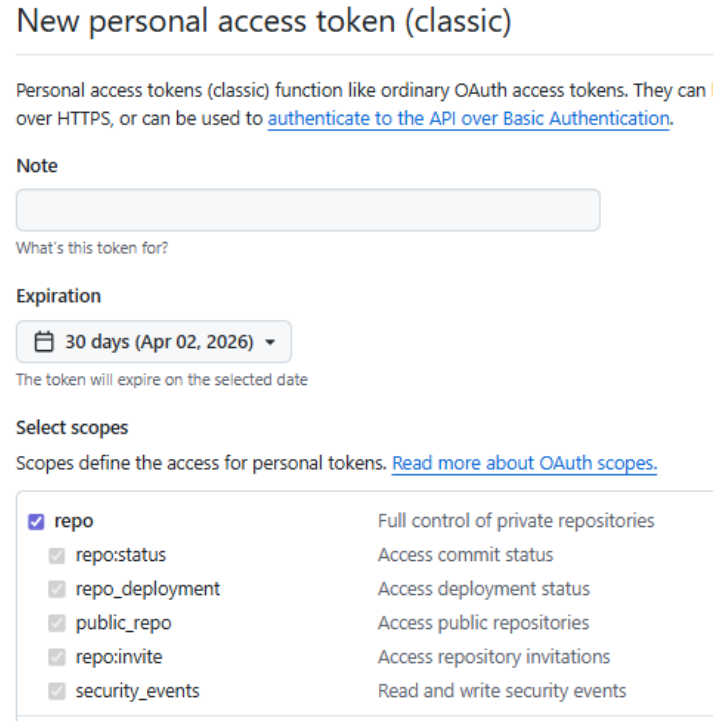
Show Git Output

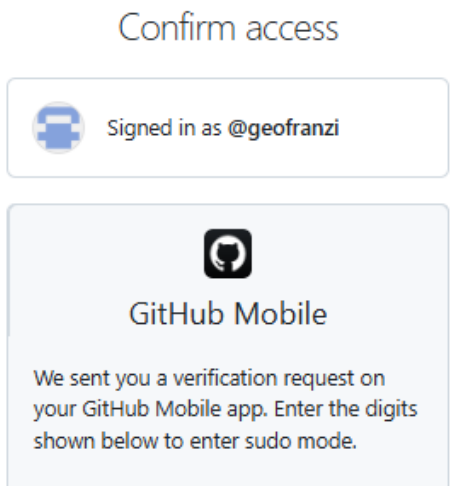
Login to GitHub and Create Token











Save the displayed token somewhere!

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Tasks

- Open command line (bash) and clone the repo (git clone https://github.com/TIPPECC/Training_2026.git)
- Navigate thru the folder structure and view content (cd / cd .., ls –l)
- Create a new text file with your name (kate NAME.txt) under "/tasks/. Add your favourite animal(s).
- Update local repo, stage ,and commit your new file with a meaningful commit message associated to the related issue und push to repo.

  git pull (update first, if there are already changes)

  git add NAME.txt

  git commit –m"commit message #3"

  git push

  -> enter username + password (=token))
- Create an GitHub issue for your neighbour describing how the animal should be described more in detail. Assign the issue. Do your own issue.

When (not) to use Git?

- ✓ Multiple people collaborate
- ✓ You need version history (Software development, Research projects, Academic writing …)
- ✓ You want to experiment safely (Testing new features, …)
- ✓ Reproducibility matters (scientific workflows, data processing pipelines, …)
- ✓ Long-term maintenance is required (research code, teaching materials, software projects…)

- Large binary files dominate
- It's a small, single-user file with no need for history
- Live databases
- Extremely large datasets (Big Data)

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Questions?

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